

Prediction-Powered Data Fusion for Treatment Effect Estimation

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Randomized controlled trials identify causal effects under transparent design assumptions, but their sample sizes can make average and conditional treatment-effect estimation imprecise. Observational studies are often larger and can yield accurate outcome predictions, but those predictions need not be causal. We develop a model-agnostic prediction-powered framework that keeps the randomized trial as the causal anchor while using observational outcome models and covariates through two tunable channels. The coefficient λ changes the outcome regression inside an RCT-valid pseudo-outcome, and ω introduces a prediction-powered control variate centered across the two samples.

For the average treatment effect, we derive exact conditional unbiasedness, a within-/between-covariate variance decomposition, the joint population variance minimizer, its population-orthogonal simplification, singular and constrained cases, and an honest plug-in algorithm. A separate conditional central limit theorem supports Wald inference after independent coefficient selection, while nested outer cross-fitting requires additional stability analysis for variance estimation. For conditional effects, we formulate a prediction-powered squared-loss learner from an RCT-valid AIPW pseudo-outcome, distinguish pseudo-outcome-noise, fixed-loss-variance, and final learner-risk oracles, and give implementable linear-sieve and frozen-neural-basis versions with honest common-reference selection. We outline an R-loss analogue only as a prospective extension requiring a separate derivation. Exact density-ratio transport covers unequal covariate marginals; classifier odds, Riesz learning, and loss-targeted balancing enter an estimated-weight regime with explicit drift. Existing simulations support the ATE construction in several small-trial regimes. CATE experiments and real-outcome analyses remain planned, so neither the theory nor the current evidence establishes universal finite-sample improvement for estimated coefficients.

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1. Introduction

1.1. Categorized literature review

Prediction-powered inference. Prediction-powered inference uses a large set of machine predictions together with a smaller labeled sample, correcting the prediction-based estimate with labeled residuals so that the statistical target does not require the predictor to be correct (Angelopoulos et al., 2023). This separation between predictive quality and inferential validity is the starting point for the present construction. Randomized observations play the role of causally valid labeled data, while observational outcome models supply potentially useful predictions.

Adaptive prediction-powered inference. PPI++ introduces data-adaptive power tuning that adjusts how strongly predictions enter an estimating equation and reports computational and statistical gains over the original construction (Angelopoulos et al., 2024). Its central lesson is that a prediction should be used in proportion to its empirical value rather than being accepted as truth. Our coefficient ω follows this control-variate logic, but the target and score are causal quantities identified by the randomized design.

Causal data fusion and transport. General causal data-fusion theory distinguishes data collected under different regimes and populations and asks whether a target causal query is identified from their combination (Bareinboim and Pearl, 2016). Reviews of causal data integration likewise separate randomized trials, observational studies, historical controls, and heterogeneous target populations (Shi et al., 2023). The present paper assumes that the trial already identifies the target effect and studies an estimation question: how can observational predictions improve precision without replacing randomized identifying information?

Combining randomized and observational estimates. Existing estimators combine experimental and observational effect estimates through shrinkage, bias detection, or adaptive weighting. Shrinkage estimators can reduce risk when observational bias is sufficiently controlled (Rosenman et al., 2023), while adaptive combination methods move toward the randomized estimator when observational bias is detected (Cheng and Cai, 2021). These approaches combine effect estimates from the two sources. Our construction instead inserts the observational outcome model into an RCT-valid pseudo-outcome and separately centers a prediction-based covariate statistic.

CATE learners and orthogonal losses. Meta-learners convert standard regression algorithms into conditional effect estimators (Künzel et al., 2019). The R-learner residualizes outcomes and treatments to isolate treatment-effect heterogeneity (Nie and Wager, 2021), while doubly robust pseudo-outcome regression can retain favorable error structure under nuisance estimation (Kennedy, 2023). Direct multi-source learners have also been proposed for heterogeneous causal data fusion (Li et al., 2022). These results motivate treating the prediction-powered construction as a loss wrapper that can be paired with a linear sieve or a fixed learned representation.

Covariate shift and balancing. When trial and observational covariate marginals differ, exact density-ratio weighting transports fixed covariate functions between populations. Riesz-representer methods provide an optimization perspective on such linear functionals and on debiasing (Chernozhukov et al., 2021). Density-ratio estimation, generic balancing, and Riesz learning are established tools. The project-specific question is whether they can be targeted to the generated CATE loss class $\{(g - t)^2 : t \in \mathcal{T}\}$ while preserving an RCT-defined risk; we treat this specialization as a working research direction rather than a settled novelty claim.

2. Background and Problem Setting

2.1. Prediction-powered inference and PPI++

Let (X_i, Y_i) , $i = 1, \dots, n$, be labeled pairs and let $\tilde{X}_j$, $j = 1, \dots, N$, be unlabeled covariates. Suppose a fixed predictor supplies $f(X_i)$ and $f(\tilde{X}_j)$. For $\omega \in \mathbb{R}$, the PPI++ estimator (Angelopoulos et al., 2024) of the labeled-population mean is

$$\hat{\mu}_{\text{PPI++}}(\omega) \triangleq \frac{1}{n} \sum_{i=1}^n Y_i + \omega \left\{ \frac{1}{N} \sum_{j=1}^N f(\tilde{X}_j) - \frac{1}{n} \sum_{i=1}^n f(X_i) \right\}. \quad (1)$$

When labeled and unlabeled covariates share a marginal law and the two samples are independent, the expression in braces has mean zero for every fixed predictor f . PPI++ selects ω to control how strongly this mean-zero prediction difference enters the estimator. Our construction retains this control-variate role and introduces a separate outcome-regression blending coefficient only after defining the RCT-valid score.

2.2. RCT-OBS problem setting and notation

Let $\mathcal{K} \triangleq \{\text{eval}, \text{nuis}, \text{tune}\}$. For each source $S \in \{R, O\}$ and role $k \in \mathcal{K}$, observe

$$D_S^k \triangleq \{V_{S,i}^k = (X_{S,i}^k, A_{S,i}^k, Y_{S,i}^k) : i = 1, \dots, n_S^k\} \sim P_S^{\otimes n_S^k}. \quad (2)$$

Here X takes values in the covariate space $\mathcal{X}$, $A \in \{0, 1\}$ is treatment, and Y is the observed outcome. The roles eval, nuis, and tune denote final evaluation, nuisance and density-ratio fitting, and coefficient or candidate tuning, respectively. Set $n \triangleq n_R^{\text{eval}}$ and $N \triangleq n_O^{\text{eval}}$, and define $\mathbb{P}_{R,n}h \triangleq n^{-1} \sum_i h(V_{R,i}^{\text{eval}})$ and $\mathbb{P}_{O,N}h \triangleq N^{-1} \sum_j h(V_{O,j}^{\text{eval}})$. Every displayed moment is finite; square-integrability and stronger moment conditions are stated locally when variance estimation or Wald inference requires them.

Let $Y(1)$ and $Y(0)$ denote potential outcomes. The RCT-population conditional and average treatment effects are

$$\tau(x) \triangleq \mathbb{E}_R\{Y(1) - Y(0) \mid X = x\}, \quad \theta \triangleq \mathbb{E}_R\{\tau(X)\}. \quad (3)$$

Assumption 1 (Trial identification). Consistency holds, $\{Y(0), Y(1)\} \perp A \mid X$ under P_R , and the known trial propensity $e(x) \triangleq P_R(A = 1 \mid X = x)$ satisfies $\epsilon < e(X) < 1 - \epsilon$ almost surely for a constant $\epsilon > 0$.

We study both the common-covariate regime $P_R^X = P_O^X$, formalized in Assumption 2, and the more general covariate-shift regime $P_R^X \neq P_O^X$, formalized in Assumption 3.

Assumption 2 (Common covariate marginal). The trial and observational covariates have the same marginal law, $P_R^X = P_O^X$. This assumption does not equate the treatment or outcome mechanisms across sources, and the observational treatment contrast may remain confounded.

Assumption 3 (Exact covariate transport). Under covariate shift, $P_R^X \ll P_O^X$ and the exact density ratio

$$r(x) \triangleq \frac{dP_R^X}{dP_O^X}(x), \quad \mathbb{E}_O\{r(X)h(X)\} = \mathbb{E}_R\{h(X)\} \quad (4)$$

is available for every required integrable function h . The common-marginal model is the special case $r = 1$.

Assumptions 2 and 3 describe alternative covariate regimes, not cumulative restrictions. Under Assumption 2, no transport weighting is needed and we set $r \triangleq 1$. When the covariate marginals differ, Assumption 3 is needed to convert OBS covariate expectations into expectations under the RCT population. Neither regime assumes that the OBS treatment contrast is causal.

Assumption 4 (Honest sample separation). The six datasets in Equation (2) are mutually independent. Nuisance and density-ratio estimators use only $D_R^{\text{nuis}}, D_O^{\text{nuis}}$. Coefficients and candidate rules use only $D_R^{\text{tune}}, D_O^{\text{tune}}$ after the nuisance objects are fixed. If candidate fitting and validation both use tuning data, their assignments are prespecified and disjoint. Final point estimation, variance estimation, and Wald inference use only $D_R^{\text{eval}}, D_O^{\text{eval}}$, after every learned object and split assignment has been fixed.

Throughout, every P_R - or P_O -expectation, covariance, variance, and exact variance result that involves a learned object is conditional on the realized nuisance and tuning datasets, split assignments, and pre-evaluation algorithmic randomness. These objects are held fixed, and only the evaluation observations remain random. We suppress this conditioning in the notation.

For $S \in \{R, O\}$ and $a \in \{0, 1\}$, define the population outcome-regression target $\mu_{S,a} : \mathcal{X} \rightarrow \mathbb{R}$ by

$$\mu_{S,a}(x) \triangleq \mathbb{E}_S[Y \mid X = x, A = a].$$

For each $S \in \{R, O\}$ and $a \in \{0, 1\}$, let $\hat{\mu}_{S,a} : \mathcal{X} \rightarrow \mathbb{R}$ denote a fixed pre-evaluation predictor of the corresponding population regression $\mu_{S,a}$. The RCT predictor is trained only on D_R^{nuis} , and the OBS predictor is trained only on D_O^{nuis} . No correctness assumption is imposed on either fitted predictor. Define

$$\hat{\mu}_R \triangleq (\hat{\mu}_{R,0}, \hat{\mu}_{R,1}), \quad \hat{\mu}_O \triangleq (\hat{\mu}_{O,0}, \hat{\mu}_{O,1}).$$

Then define the observational treatment-effect prediction

$$g(x) \triangleq \hat{\mu}_{O,1}(x) - \hat{\mu}_{O,0}(x), \quad (5)$$

Definition 1 (RCT-valid AIPW score). Let $q \triangleq (q_0, q_1)$ be a fixed pair of measurable functions from $\mathcal{X}$ to $\mathbb{R}$. Define

$$\varphi(V; q) \triangleq q_1(X) - q_0(X) + \frac{A}{e(X)}\{Y - q_1(X)\} - \frac{1 - A}{1 - e(X)}\{Y - q_0(X)\}. \quad (6)$$

Proposition 1 (RCT-validity for an arbitrary outcome regression). Under Assumptions 1 and 4, every fixed q with finite first moments satisfies

$$\mathbb{E}_R\{\varphi(V; q) \mid X\} = \tau(X). \quad (7)$$

Definition 2 (Fusion score). For $\lambda \in \mathbb{R}$, define

$$\hat{\mu}_\lambda \triangleq (1 - \lambda)\hat{\mu}_R + \lambda\hat{\mu}_O, \quad Z_\lambda \triangleq \varphi(V; \hat{\mu}_\lambda) = Z_0 + \lambda\Delta, \quad \Delta \triangleq Z_1 - Z_0. \quad (8)$$

Equation (7) implies $\mathbb{E}_R(Z_\lambda \mid X) = \tau(X)$ and $\mathbb{E}_R(\Delta \mid X) = 0$. The observational contrast g is used only as a prediction and need not equal τ .

Definition 3 (RCT-only AIPW estimator). Let $\mu_R \triangleq (\mu_{R,0}, \mu_{R,1})$. Setting $q = \mu_R$ in Definition 1 gives the oracle RCT AIPW score. Its nuisance-sample fitted counterpart is $Z_0 = \varphi(V; \hat{\mu}_R)$. Define the RCT-only AIPW estimator on the evaluation sample by

$$\hat{\theta}_{\text{AIPW},R} \triangleq \mathbb{P}_{R,n}\varphi(V; \hat{\mu}_R) = \mathbb{P}_{R,n}Z_0.$$

Corollary 1 (RCT-only AIPW recovery). Under Assumptions 1 and 4, $\hat{\theta}_{\text{AIPW},R}$ is unbiased for θ over D_R^{eval} .

2.3. Existing data-fusion estimators

Three existing strategies provide the nearest comparisons. First, shrinkage methods combine an unbiased experimental estimate with a potentially biased observational estimate and trade variance against observational bias (Rosenman et al., 2023). Second, adaptive RCT-OBS estimators estimate source-specific conditional effects and downweight the observational estimator when its discrepancy from the trial is large (Cheng and Cai, 2021). Third, direct multi-source learners construct weighted pseudo-outcomes for heterogeneous effects across sources (Li et al., 2022). Our estimator differs at the level of the fusion object: λ changes the nuisance regression inside a trial-valid score, while ω centers a fixed prediction across study covariate samples. Neither channel treats the observational treatment contrast as identified causal truth.

Within the adaptive and integrative line, the elastic procedure of Yang et al. (2023) tests whether real-world data are sufficiently compatible with the randomized trial before pooling information for prespecified effect modifiers. The confounding-function approach of Yang et al. (2025) instead introduces and jointly estimates an HTE function and an observational confounding function under explicit cross-source identification conditions. These methods use observational outcomes through a poolability decision or an identified confounding-function model; our two fusion channels retain the RCT-valid score as the causal anchor.

3. Prediction-Powered Fusion for the ATE

3.1. Estimator and exact variance

For fixed (λ, ω) , define

$$\hat{\theta}_r(\lambda, \omega) = \mathbb{P}_{R,n}Z_\lambda + \omega \{ \mathbb{P}_{O,N}\{r(X)g(X)\} - \mathbb{P}_{R,n}g(X) \}. \quad (9)$$

Here Z_λ is the fusion score in Definition 2, g is defined in Equation (5), and r is the transport ratio in Assumption 3. At the coefficient origin,

$$\hat{\theta}_r(0, 0) = \mathbb{P}_{R,n}Z_0 = \hat{\theta}_{\text{AIPW},R},$$

so the proposed family contains the RCT-only AIPW estimator from Definition 3 exactly. The first term averages an RCT-valid pseudo-outcome. Under Assumption 3, the expression in braces is a mean-zero control variate; under Assumption 2, this is the special case $r = 1$. Thus, the RCT identifies the target, λ changes within-covariate score noise, and ω uses the large OBS covariate sample to remove predictable between-covariate variation.

Theorem 1 (Exact mean and variance under honest evaluation). Under Assumptions 1, 3, and 4, suppose $n, N > 0$ and the displayed scores are square-integrable. Every fixed (λ, ω) then satisfies

$$\mathbb{E}\{\hat{\theta}_r(\lambda, \omega)\} = \theta, \quad (10)$$

$$\text{Var}\{\hat{\theta}_r(\lambda, \omega)\} = \frac{1}{n} \text{Var}_R\{Z_\lambda - \omega g(X)\} + \frac{\omega^2}{N} \text{Var}_O\{r(X)g(X)\}. \quad (11)$$

Equivalently, the law of total variance gives the within-/between-covariate decomposition

$$\text{Var}\{\hat{\theta}_r(\lambda, \omega)\} = \frac{1}{n} \mathbb{E}_R\{\text{Var}_R(Z_\lambda \mid X)\} + \frac{1}{n} \text{Var}_R\{\tau(X) - \omega g(X)\} + \frac{\omega^2}{N} \text{Var}_O\{r(X)g(X)\}. \quad (12)$$

The mean follows from $\mathbb{E}_R(Z_\lambda) = \theta$ and $\mathbb{E}_O(rg) = \mathbb{E}_R(g)$. The variance follows after rewriting Equation (9) as the sum of independent RCT and OBS empirical means.

Corollary 2 (Common-marginal specialization). Under Assumptions 1, 2, and 4, set $r = 1$ in Theorem 1. Then Equation (9) is exactly unbiased, and its OBS variance term is $\omega^2 \text{Var}_R\{g(X)\}/N$.

3.2. Joint and population-orthogonal oracles

Throughout this subsection, Assumption 4 remains in force, n and N are positive, and the displayed random variables have finite second moments.

Definition 4 (ATE variance-quadratic objects). Define the five population moments

$$A \triangleq \text{Var}_R(\Delta), \quad B \triangleq \text{Var}_R(g) + \frac{n}{N} \text{Var}_O(rg), \quad (13)$$

$$C \triangleq \text{Cov}_R(Z_0, \Delta), \quad D \triangleq \text{Cov}_R(Z_0, g), \quad E \triangleq \text{Cov}_R(\Delta, g). \quad (14)$$

The moment A is the curvature cost of changing the outcome-regression blend λ . The moment B is the curvature cost of the prediction-powered scale ω , including both the RCT subtraction and the scaled OBS addition. The moment C is the baseline-score covariance that determines the linear slope in the λ direction. The moment D is the baseline-prediction covariance that determines the linear slope in the ω direction, with a minus sign because the RCT score contains $-\omega g$. The moment E is the interaction covariance coupling the two coefficients through the cross-term $-2E\lambda\omega$.

Collect these moments and coefficients as

$$\mathbf{H}_{\text{ATE}} \triangleq \begin{pmatrix} A & -E \\ -E & B \end{pmatrix}, \quad \boldsymbol{\ell} \triangleq \begin{pmatrix} C \\ -D \end{pmatrix}, \quad \boldsymbol{\beta} \triangleq \begin{pmatrix} \lambda \\ \omega \end{pmatrix}, \quad (15)$$

Proposition 2 (ATE variance quadratic and positive-semidefinite curvature). Under the standing conditions of this subsection, the compact variance in Equation (11) is

$$\text{Var}\{\widehat{\theta}_r(\boldsymbol{\beta})\} = \frac{1}{n} \left\{ \text{Var}_R(Z_0) + 2\boldsymbol{\ell}^\top \boldsymbol{\beta} + \boldsymbol{\beta}^\top \mathbf{H}_{\text{ATE}} \boldsymbol{\beta} \right\}. \quad (16)$$

Moreover, $\mathbf{H}_{\text{ATE}}$ is positive semidefinite because, for every $(u, v) \in \mathbb{R}^2$,

$$\begin{pmatrix} u & v \end{pmatrix} \mathbf{H}_{\text{ATE}} \begin{pmatrix} u \\ v \end{pmatrix} = \text{Var}_R(u\Delta - vg) + \frac{n}{N} v^2 \text{Var}_O(rg) \geq 0. \quad (17)$$

The variance in Equation (16) contains the RCT-only baseline $\text{Var}_R(Z_0)/n$, which no coefficient can change. Coefficient selection can therefore focus on the remaining quadratic. Its positive-semidefinite curvature guarantees that this reduced objective cannot decrease without bound.

Definition 5 (ATE coefficient objective). Define the coefficient-dependent part of the scaled variance by

$$\mathcal{J}_{\text{ATE}}(\boldsymbol{\beta}) \triangleq 2\boldsymbol{\ell}^\top \boldsymbol{\beta} + \boldsymbol{\beta}^\top \mathbf{H}_{\text{ATE}} \boldsymbol{\beta}.$$

Proposition 3 (Population orthogonality of the fusion directions). Under Assumptions 1 and 4, the two endpoint scores have the same conditional target:

$$\begin{aligned} \mathbb{E}_R(Z_a \mid X) &= \tau(X), & a \in \{0, 1\}, \\ \mathbb{E}_R(\Delta \mid X) &= 0. \end{aligned}$$

Because g is a fixed function of X , it follows that

$$E = \text{Cov}_R\{\Delta, g(X)\} = 0, \quad D = \text{Cov}_R\{\tau(X), g(X)\}. \quad (18)$$

This result formalizes the purpose of the outcome-regression blend: changing λ changes pseudo-outcome noise without changing its conditional CATE target. The zero population interaction separates the two oracle directions. We nevertheless retain the general E term because its tuning-sample analogue $\hat{E}$ is generally nonzero, and the empirical joint quadratic program must account for that sampling fluctuation.

Proposition 4 (Equivalent ATE oracle objectives). For every nonempty feasible set $\mathcal{B} \subseteq \mathbb{R}^2$, the two argmin sets, possibly empty when the infimum is not attained, satisfy

$$\arg \min_{\beta \in \mathcal{B}} \text{Var}\{\hat{\theta}_r(\beta)\} = \arg \min_{\beta \in \mathcal{B}} \mathcal{J}_{\text{ATE}}(\beta).$$

Positive semidefiniteness means that the variance objective cannot curve downward along any direction (u, v) . For this two-dimensional matrix, the stronger condition $AB - E^2 > 0$, together with positive semidefiniteness, makes $\mathbf{H}_{\text{ATE}}$ positive definite and invertible. It turns a possibly flat quadratic into one with a unique unrestricted minimizer.

Definition 6 (Joint and ridge population oracles). When $AB - E^2 > 0$, define the unrestricted joint oracle by

$$\beta_{\text{joint}}^* \triangleq \arg \min_{\beta \in \mathbb{R}^2} \text{Var}\{\hat{\theta}_r(\beta)\} = \arg \min_{\beta \in \mathbb{R}^2} \mathcal{J}_{\text{ATE}}(\beta).$$

The equality of the two argmin sets follows from Proposition 4. For a prespecified $\rho > 0$, define the ridge-stabilized population oracle by

$$\beta_{\rho}^* \triangleq \arg \min_{\beta \in \mathbb{R}^2} \left\{ \mathcal{J}_{\text{ATE}}(\beta) + \rho \|\beta\|_2^2 \right\}.$$

Let I_2 denote the 2×2 identity matrix.

Theorem 2 (Joint and ridge population oracles). If $AB - E^2 > 0$, the joint oracle in Definition 6 is unique and

$$\beta_{\text{joint}}^* = \begin{pmatrix} \lambda_{\text{joint}}^* \\ \omega_{\text{joint}}^* \end{pmatrix} = -\mathbf{H}_{\text{ATE}}^{-1} \ell = \begin{pmatrix} (ED - BC)/(AB - E^2) \\ (AD - EC)/(AB - E^2) \end{pmatrix}. \quad (19)$$

For every $\rho > 0$, whether or not $\mathbf{H}_{\text{ATE}}$ is singular, the ridge oracle is unique and

$$\beta_{\rho}^* = -(\mathbf{H}_{\text{ATE}} + \rho I_2)^{-1} \ell. \quad (20)$$

Theorem 2 locates the joint minimum of the two-coefficient variance quadratic. Corollary 3 answers the distinct comparison question: it evaluates the quadratic at that minimum and quantifies the improvement relative to the RCT-only origin $\beta = 0$. The general interaction E is retained here because its empirical analogue $\hat{E}$ need not vanish, even though the exact population model later implies $E = 0$.

Remark 1 (Singular or constrained curvature). If $\mathbf{H}_{\text{ATE}}$ is singular, an unpenalized joint minimum can lie along a flat set rather than at a unique point. The Moore–Penrose and null-space characterization, as well as existence and uniqueness results under coefficient constraints, are recorded in Appendix B. The ridge oracle remains unique for every $\rho > 0$. In implementation we minimize the continuous quadratic over a prespecified compact box, which guarantees at least one constrained solution.

Corollary 3 (Joint-oracle variance reduction). Let $\sigma_0^2 = \text{Var}_R(Z_0)$ and $\mathcal{V}_0 = \sigma_0^2/n$ be the variance at $(\lambda, \omega) = (0, 0)$. If $AB - E^2 > 0$, the unrestricted joint oracle has

$$V_{\min} = \mathcal{V}_0 - \frac{1}{n} \frac{BC^2 - 2ECD + AD^2}{AB - E^2},$$

$$\mathcal{V}_0 - V_{\min} = \frac{1}{n} \boldsymbol{\ell}^\top \mathbf{H}_{\text{ATE}}^{-1} \boldsymbol{\ell}.$$

The reduction is nonnegative and is strictly positive exactly when $\boldsymbol{\ell} \neq 0$, equivalently when $(C, D) \neq (0, 0)$. Under population orthogonality $E = 0$ and $A, B > 0$, this specializes to

$$V_{\min} = \mathcal{V}_0 - \frac{1}{n} \left(\frac{C^2}{A} + \frac{D^2}{B} \right).$$

By Proposition 3, the interaction term $-2E\lambda\omega$ vanishes. When $A, B > 0$, the population oracle therefore separates:

$$\lambda^* = -\frac{C}{A}, \quad \omega^* = \frac{D}{B}. \quad (21)$$

If $A = 0$, conditional mean zero forces $\Delta = 0$ almost surely and the λ direction is flat. If $B = 0$, both $g(X)$ and $r(X)g(X)$ are constant on their relevant supports, so the prediction-powered correction is samplewise zero and the ω direction is flat. Under the common covariate marginal,

$$B = \left(1 + \frac{n}{N} \right) \text{Var}_R\{g(X)\}.$$

For the rectangular box $\mathcal{B}_\beta$ and positive A, B , population orthogonality gives the coordinatewise projection

$$\lambda_{\mathcal{B}_\beta}^* = \Pi_{[L_\lambda, U_\lambda]}(-C/A), \quad \omega_{\mathcal{B}_\beta}^* = \Pi_{[L_\omega, U_\omega]}(D/B).$$

For a coupled constraint or a nonzero interaction, the original quadratic must be minimized directly. The population oracle weakly improves the variance objective whenever $(0, 0)$ is feasible. Estimated coefficients have no universal finite-sample no-harm guarantee.

3.3. Estimable honest algorithm

For random quantities U and W evaluated on D_S^{tune} , assume $n_S^{\text{tune}} \geq 2$. Conditional on the fixed nuisance fits, let $\widehat{\text{Cov}}_{S, \text{tune}}(U, W)$ denote their usual Bessel-corrected sample covariance and define $\widehat{\text{Var}}_{S, \text{tune}}(U) \triangleq \widehat{\text{Cov}}_{S, \text{tune}}(U, U)$. The factor n/N below uses the evaluation-sample sizes whose

variance is optimized.

Using the tuning samples, compute

$$\hat{A} = \widehat{\text{Var}}_{R,\text{tune}}(\Delta), \quad \hat{B} = \widehat{\text{Var}}_{R,\text{tune}}(g) + \frac{n}{N} \widehat{\text{Var}}_{O,\text{tune}}(rg), \quad (22)$$

$$\hat{C} = \widehat{\text{Cov}}_{R,\text{tune}}(Z_0, \Delta), \quad \hat{D} = \widehat{\text{Cov}}_{R,\text{tune}}(Z_0, g), \quad \hat{E} = \widehat{\text{Cov}}_{R,\text{tune}}(\Delta, g). \quad (23)$$

Define

$$\hat{\mathbf{H}}_{\text{ATE}} = \begin{pmatrix} \hat{A} & -\hat{E} \\ -\hat{E} & \hat{B} \end{pmatrix}, \quad \hat{\ell} = \begin{pmatrix} \hat{C} \\ -\hat{D} \end{pmatrix}, \quad \hat{d} = \hat{A}\hat{B} - \hat{E}^2.$$

Choose positive thresholds t_A, t_B, t_{det} before observing the tuning outcomes. The population-orthogonality axis rule is

$$\hat{\lambda}_{\text{axis}} = \begin{cases} -\hat{C}/\hat{A}, & \hat{A} > t_A, \\ 0, & \hat{A} \leq t_A, \end{cases} \quad \hat{\omega}_{\text{axis}} = \begin{cases} \hat{D}/\hat{B}, & \hat{B} > t_B, \\ 0, & \hat{B} \leq t_B. \end{cases}$$

When $\hat{A} > t_A$, $\hat{B} > t_B$, and $\hat{d} > t_{\text{det}}$, the unrestricted empirical joint rule is

$$\hat{\lambda}_{\text{joint}} = \frac{\hat{E}\hat{D} - \hat{B}\hat{C}}{\hat{d}}, \quad \hat{\omega}_{\text{joint}} = \frac{\hat{A}\hat{D} - \hat{E}\hat{C}}{\hat{d}}.$$

For the prespecified nonempty compact box $\mathcal{B}_\beta = [L_\lambda, U_\lambda] \times [L_\omega, U_\omega]$ containing the origin, the boundary-correct empirical choice is

$$\hat{\beta}_{\mathcal{B}_\beta} \in \arg \min_{\beta \in \mathcal{B}_\beta} \left\{ 2\hat{\ell}^\top \beta + \beta^\top \hat{\mathbf{H}}_{\text{ATE}} \beta \right\}.$$

If the unrestricted joint rule lies outside $\mathcal{B}_\beta$, this quadratic program must be solved rather than clipping its coordinates separately; rectangular coordinate-wise clipping is exact only after imposing $\hat{E} = 0$. For the joint rule, if a required threshold fails or the quadratic solver is numerically unstable, the prespecified fallback is $(0, 0)$; the axis rule uses its coordinate-specific threshold definitions above.

The population-orthogonality plug-in imposes $\hat{E} = 0$ and uses $(-\hat{C}/\hat{A}, \hat{D}/\hat{B})$. The empirical joint minimizer retains $\hat{E}$ and applies Equation (19). The two rules differ because sample orthogonality is not exact.

Each sample covariance is conditionally unbiased for its corresponding fixed-score population moment given the nuisance fits, but the ratios defining $\hat{\lambda}$ and $\hat{\omega}$ are nonlinear and are not generally unbiased for the population oracle coefficients. The thresholds and feasible box control instability; they do not turn these selected coefficients into finite-sample oracle estimates.

The displayed analytic ratios apply only when their corresponding denominator thresholds pass. Imposing $\hat{E} = 0$ does not assert that the realized tuning covariance vanishes. It applies the proven population restriction $E = 0$ before solving the sample problem. Retaining $\hat{E} = \widehat{\text{Cov}}_{R,\text{tune}}(\Delta, g)$ instead minimizes the realized empirical quadratic, whose interaction term is generally nonzero because of tuning-sample noise.

Algorithm 1 distinguishes the independent-role implementation covered by the exact finite-sample

results from optional nested cross-fitting. Independent role samples may be supplied directly; physically splitting one dataset is only one way to construct them.

Algorithm 1: ATE estimation with honest evaluation and optional cross-fitting

Input: Either role-indexed samples satisfying Assumption 4 or prespecified paired outer folds; a feasible box and denominator thresholds.

- 1 **Independent-role implementation:** Fit $\hat{\mu}_R, \hat{\mu}_O, g$, and any ratio model on $D_R^{\text{nuis}}, D_O^{\text{nuis}}$. Compute the five moments and select the coefficients on $D_R^{\text{tune}}, D_O^{\text{tune}}$. Evaluate Equation (9) on $D_R^{\text{eval}}, D_O^{\text{eval}}$. The exact theorem uses the true r ; replacing it by $\hat{r}$ invokes the drift conditions below.
- 2 **Optional cross-fitted implementation:** For each outer fold k , fit nuisances and tune coefficients without that fold, score only its held-out RCT and OBS observations, and aggregate $\hat{\theta}_{\text{CF}} = \sum_{k=1}^K a_k \hat{\theta}_k$ using fixed $a_k \geq 0$ with $\sum_k a_k = 1$. Use $a_k = 1/K$ for equal-sized folds.
- 3 **Infer under independent roles:** Set $\psi_R = Z_{\hat{\lambda}} - \hat{\omega}g$ and $\psi_O = \hat{\omega}rg$, and report $\hat{V} = \widehat{\text{Var}}_{R,\text{eval}}(\psi_R)/n + \widehat{\text{Var}}_{O,\text{eval}}(\psi_O)/N$. Under conditional Lindeberg conditions, bounded conditional $(2 + \delta)$ moments, and a nondegenerate variance limit, use $\hat{\theta}_r \pm z_{1-\alpha/2} \sqrt{\hat{V}}$.
- 4 **Cross-fitted inference caveat:** Do not sum foldwise variances or report a pooled analytic Wald interval without a separate stability or influence-function argument.

Output: Independent roles: $\hat{\theta}_r, \hat{V}$, and its Wald interval; optional cross-fitting: $\hat{\theta}_{\text{CF}}$ with the stated inference caveat.

For explicit evaluation notation, define

$$\psi_{R,i} = Z_{\hat{\lambda}}(V_{R,i}^{\text{eval}}) - \hat{\omega}g(X_{R,i}^{\text{eval}}), \quad \psi_{O,j} = \hat{\omega}r(X_{O,j}^{\text{eval}})g(X_{O,j}^{\text{eval}}),$$

and let $\widehat{\text{Var}}_{R,\text{eval}}$ and $\widehat{\text{Var}}_{O,\text{eval}}$ be the usual unbiased sample variances with denominators $n - 1$ and $N - 1$, respectively. For $n, N \geq 2$, the fixed-selected-coefficient variance estimator and Wald interval are

$$\hat{V} = \frac{\widehat{\text{Var}}_{R,\text{eval}}(\psi_R)}{n} + \frac{\widehat{\text{Var}}_{O,\text{eval}}(\psi_O)}{N},$$

$$\hat{\theta}_r \pm z_{1-\alpha/2} \sqrt{\hat{V}}.$$

Unbiasedness of $\hat{V}$ follows from Assumption 4. Wald coverage additionally requires the stated Lindeberg, moment, and nondegeneracy conditions; it is not a finite-sample coverage statement. Here $z_{1-\alpha/2}$ is the $(1 - \alpha/2)$ -quantile of the standard normal distribution; for $\alpha = 0.05$, it is approximately 1.96.

Proposition 5 (Exact variance estimation after honest selection). Under Theorem 1, let $(\hat{\lambda}, \hat{\omega})$ be selected without the evaluation samples, as required by Assumption 4, and suppose $n, N \geq 2$. The selected estimator is unbiased and

$$\mathbb{E}(\hat{V}) = \text{Var}\{\hat{\theta}_r(\hat{\lambda}, \hat{\omega})\}.$$

Theorem 3 (Final-sample central limit theorem and Wald inference). In addition to Proposition 5, suppose $n, N \rightarrow \infty$, $n/N \rightarrow \kappa \in [0, \infty)$, and for some $\delta > 0$ the centered RCT and OBS evaluation-score arrays have uniformly bounded $(2 + \delta)$ moments after the learned objects are fixed. This moment assumption supplies the required Lindeberg condition. Assume also that

$$n \operatorname{Var}\{\hat{\theta}_r(\hat{\lambda}, \hat{\omega})\} \rightarrow \sigma^2 \in (0, \infty)$$

in probability. Then

$$\frac{\hat{\theta}_r - \theta}{\sqrt{\hat{V}}} \rightsquigarrow N(0, 1),$$

and the displayed Wald interval has asymptotic coverage $1 - \alpha$. Variance unbiasedness alone does not imply this coverage statement.

With an estimated ratio $\hat{r}$, define

$$h_n(X) = \{\hat{r}(X) - r(X)\}g(X). \quad (24)$$

The exact evaluation-sample drift is

$$\omega b_{\hat{r}} \triangleq \omega \mathbb{E}_O\{h_n(X)\}. \quad (25)$$

Honest separation prevents evaluation leakage but does not remove this drift. A direct sufficient negligible-error regime is

$$\sqrt{n} \omega \mathbb{E}_O\{h_n(X)\} = o_p(1), \quad |\omega| \sqrt{\frac{n}{N}} \|h_n - \mathbb{E}_O(h_n)\|_{L^2(P_O)} = o_p(1). \quad (26)$$

Evaluation-only variance is first-order adequate under these conditions and the preceding CLT assumptions. Otherwise the ratio learner’s influence-function contribution, the OBS empirical-process term, and their covariances must be included.

Estimating the density ratio. All three constructions below use only the nuisance samples and are fixed before coefficient tuning and final evaluation. Let $\mathbb{P}_R^{\text{nuis}}$ and $\mathbb{P}_O^{\text{nuis}}$ denote the corresponding empirical averages. First, pool the nuisance-sample covariates and let $S = 1$ indicate an RCT observation and $S = 0$ an OBS observation. With $\pi = P(S = 1)$ and $s(x) = P(S = 1 \mid X = x)$, Bayes’ rule gives

$$r(x) = \frac{1 - \pi}{\pi} \frac{s(x)}{1 - s(x)}, \quad \hat{r}_{\text{cls}}(x) = \frac{1 - \hat{\pi}}{\hat{\pi}} \frac{\hat{s}(x)}{1 - \hat{s}(x)}.$$

For an unweighted pooled classifier, $\hat{\pi} = n_R^{\text{nuis}} / (n_R^{\text{nuis}} + n_O^{\text{nuis}})$. Class weighting or resampling requires replacing this factor by the effective training prior. The odds representation also requires overlap, $0 < s(X) < 1$, on the RCT covariate support. Probability clipping stabilizes the estimate but changes it from the exact density ratio.

A Riesz construction estimates the transport functional directly (Chernozhukov et al., 2021, 2022).

Let $\phi : \mathcal{X} \rightarrow \mathbb{R}^p$ be a fixed feature map and define

$$\widehat{M}_O = \mathbb{P}_O^{\text{nuis}} \{\phi(X)\phi(X)^\top\}, \quad \widehat{b}_R = \mathbb{P}_R^{\text{nuis}} \{\phi(X)\}.$$

For $\rho > 0$, the empirical Riesz quadratic and its closed-form solution are

$$\begin{aligned} \widehat{\eta}_\rho &= \arg \min_{\eta \in \mathbb{R}^p} \left[\eta^\top \widehat{M}_O \eta - 2\widehat{b}_R^\top \eta + \rho \|\eta\|_2^2 \right] = (\widehat{M}_O + \rho I_p)^{-1} \widehat{b}_R, \\ \widehat{r}_{\text{Riesz}}(x) &= \phi(x)^\top \widehat{\eta}_\rho. \end{aligned}$$

At $\rho = 0$, a compatible singular system uses $\widehat{M}_O^+ \widehat{b}_R$. A restricted feature map estimates a projected representer, and the resulting weights need not be nonnegative.

A nonnegative automatic-balancing alternative starts with an out-of-sample-evaluable family $w_\eta : \mathcal{X} \rightarrow [0, \infty)$. For a fixed feature map ϕ , a positive semidefinite matrix W , and $\gamma > 0$, define $\|v\|_W^2 = v^\top W v$ and solve

$$\begin{aligned} \widehat{\eta}_{\text{bal}} &\in \arg \min_{\eta: \mathbb{P}_O^{\text{nuis}} w_\eta = 1} \left[\left\| \mathbb{P}_O^{\text{nuis}} \{w_\eta(X)\phi(X)\} - \mathbb{P}_R^{\text{nuis}} \{\phi(X)\} \right\|_W^2 + \gamma \mathbb{P}_O^{\text{nuis}} \{w_\eta(X)^2\} \right], \\ \widehat{r}_{\text{bal}}(x) &= w_{\widehat{\eta}_{\text{bal}}}(x). \end{aligned}$$

Including g among the coordinates of ϕ directly targets the transported moment used by the ATE correction. The dispersion penalty controls unstable weights but generally prevents exact empirical balance.

These constructions do not automatically inherit the exact-r theorem. For any fixed evaluation-independent weight function w , replacing r by w gives

$$\begin{aligned} \mathbb{E}\{\widehat{\theta}_w(\lambda, \omega)\} &= \theta + \omega \{P_O(wg) - P_R(g)\} \\ &= \theta + \omega P_O\{(w - r)g\}. \end{aligned}$$

Thus exact unbiasedness holds for $w = r$, or more generally for a fixed w satisfying the exact population balance condition $P_O(wg) = P_R(g)$. Classifier, restricted Riesz, and automatic-balancing estimates are otherwise in the estimated-ratio regime governed by the drift and rate conditions above. The Auto-DML citation motivates the Riesz learning criterion; it does not by itself make the present score Neyman-orthogonal to ratio error or supply the ratio learner's first-order variance contribution.

Nested outer cross-fitting. As an optional alternative to independent role samples, observations can be reused without scoring any observation with a nuisance fit trained on its own outcome. Prespecify paired outer folds in both sources, or randomize the folds independently and hold the realized assignments fixed. Within each outer complement, use inner splits to fit nuisances and select $(\widehat{\lambda}_k, \widehat{\omega}_k)$; then score only the untouched paired folds. With fixed nonnegative weights a_k summing to one, define

$$\widehat{\theta}_k = \mathbb{P}_{R,k} Z_{\widehat{\lambda}_k}^{(-k)} + \widehat{\omega}_k \left\{ \mathbb{P}_{O,k}(r g^{(-k)}) - \mathbb{P}_{R,k} g^{(-k)} \right\}, \quad \widehat{\theta}_{\text{CF}} = \sum_k a_k \widehat{\theta}_k.$$

Proposition 6 (What outer-fold honesty proves). Under Assumption 1 and either the common covariate marginal or exact r , suppose the outer-fold assignment is prespecified and data-independent, or independently randomized with its realized randomness held fixed. Each $\hat{\theta}_k$ is then conditionally unbiased given its own outer complement, and $\hat{\theta}_{\text{CF}}$ is unconditionally unbiased. Overlapping outer training complements make the fold estimators dependent, so these facts do not justify adding foldwise variances or using a naive pooled Wald interval.

A pooled cross-fitted interval requires an additional stability or influence-function argument, for example convergence of fold-specific RCT and OBS scores to common deterministic limits in their respective L_2 norms, uniform $(2 + \delta)$ moments, and a nondegenerate two-sample limit. The present theory does not assert a general cross-fitted variance theorem without such conditions.

Diagnostics. An implementation should report split assignments, propensity bounds, candidate boxes, all denominator and determinant thresholds, coefficient fallbacks, ratio overlap and clipping, and the selected RCT-only comparator. Near-zero $\hat{A}$, $\hat{B}$, or $\hat{d}$ must trigger the prespecified rule rather than an unstable analytic ratio.

4. Prediction-Powered Fusion Learning for CATE

4.1. Prediction-powered loss, coefficient roles, and the ATE relation

Throughout this subsection, nuisances and ratios are fit on $D_R^{\text{nuis}}, D_O^{\text{nuis}}$. Candidate fitting and honest validation use prespecified disjoint portions of $D_R^{\text{tune}}, D_O^{\text{tune}}$. Every selected object is fixed before final evaluation on $D_R^{\text{eval}}, D_O^{\text{eval}}$, as required by Assumption 4.

For a fixed candidate CATE function t , define the density-ratio-weighted prediction-powered squared loss

$$\hat{L}_{\text{DR}}(t; \lambda, \omega) = \mathbb{P}_{R,n}\{Z_\lambda - t(X)\}^2 \quad (27)$$

$$+ \omega \left[\mathbb{P}_{O,N}\{r(X)(g(X) - t(X))^2\} - \mathbb{P}_{R,n}\{(g(X) - t(X))^2\} \right]. \quad (28)$$

For every fixed candidate considered below, assume $Z_\lambda, t \in L^2(P_R)$, and let $G_t(X) = \{g(X) - t(X)\}^2$ satisfy $G_t \in L^1(P_R)$ and $rG_t \in L^1(P_O)$.

Proposition 7 (Exact target of the prediction-powered CATE loss). Under Assumptions 1, 3, and 4,

$$\mathbb{E}\{\hat{L}_{\text{DR}}(t; \lambda, \omega)\} = Q(\lambda) + \mathbb{E}_R\{t(X) - \tau(X)\}^2, \quad (29)$$

$$Q(\lambda) \triangleq \mathbb{E}_R\{Z_\lambda - \tau(X)\}^2. \quad (30)$$

The common-marginal case sets $r = 1$. In both cases, the expected loss is flat in ω .

Because $Z_\lambda = Z_0 + \lambda\Delta$ and $\mathbb{E}_R(\Delta | X) = 0$,

$$Q(\lambda) = Q(0) + 2C\lambda + A\lambda^2.$$

When $A > 0$, define the integrated pseudo-outcome-noise oracle

$$\lambda_Q^* = \arg \min_{\lambda \in \mathbb{R}} Q(\lambda) = -C/A. \quad (31)$$

Write $\lambda_{\text{ATE}}^* := \lambda^*$ and $\omega_{\text{ATE}}^* := \omega^*$ for the two coordinates of the population-orthogonal ATE variance oracle in Equation (21). Then $\lambda_Q^* = \lambda_{\text{ATE}}^*$. When $A = 0$, $\Delta = 0$ almost surely and $Q(\lambda)$ is flat in λ . The ATE variance coefficient $\omega_{\text{ATE}}^* = D/B$ is a possible training default, but Equation (30) does not identify it as a CATE-risk optimizer.

Proposition 8 (Fixed measurable and pointwise noise coefficients). Under Assumptions 1 and 4, let $\lambda(\cdot)$ be a fixed measurable function such that $Z_0 + \lambda(X)\Delta$ is integrable. Then

$$\mathbb{E}_R\{Z_0 + \lambda(X)\Delta \mid X\} = \tau(X).$$

For the pointwise statement, choose measurable versions of the displayed conditional covariance and variance and assume they are finite. For P_R^X -almost every x on which $\text{Var}_R(\Delta \mid X = x) > 0$, the conditional-noise minimizer is

$$\lambda_Q^*(x) = -\frac{\text{Cov}_R(Z_0, \Delta \mid X = x)}{\text{Var}_R(\Delta \mid X = x)}.$$

On the zero-denominator set the conditional criterion is flat, and any finite measurable value may be assigned. The ratio must be measurable on the positive-denominator set, and the resulting pseudo-outcome must remain integrable.

The global scalar λ_Q^* , the pointwise function $\lambda_Q^*(x)$, and the coefficient chosen by a restricted or penalized learner are different oracle objects.

The sampling variance of a fixed candidate loss identifies a distinct coefficient. Let $U_{t,\lambda} = \{Z_\lambda - t(X)\}^2$.

Proposition 9 (Fixed-loss variance oracle). Under Assumption 4, fix t , λ , g , and r using the nuisance and tuning samples, and evaluate the displayed loss on the independent evaluation samples. Suppose $U_{t,\lambda} \in L^2(P_R)$, $G_t \in L^2(P_R)$, and $rG_t \in L^2(P_O)$. If the denominator below is positive, the unrestricted variance-minimizing coefficient is

$$\omega_{\text{LV}}^*(t; \lambda) = \frac{\text{Cov}_R(U_{t,\lambda}, G_t)}{\text{Var}_R(G_t) + (n/N) \text{Var}_O(rG_t)}. \quad (32)$$

Under the common covariate marginal this denominator is $(1 + n/N) \text{Var}_R(G_t)$. If the denominator is zero, the empirical correction is constant and its variance criterion is flat in ω .

This oracle minimizes the variance of one fixed empirical loss. It does not minimize the final learned CATE risk. The square-integrability conditions are fourth-moment-type requirements on the underlying residuals.

Final learner-risk oracle. Let $\mathfrak{C}$ be a prespecified finite candidate class. An index

$$c = (\lambda_c, \omega_c, p_c, \rho_c) \in \mathfrak{C} \quad (33)$$

specifies the two fusion coefficients, the sieve dimension, and the penalty level. All other components of the learning rule, including the role-sample allocation, nuisance algorithms, basis construction, and optimization convention, are also prespecified. For realized nuisance and candidate-fitting samples, let $\hat{t}_c$ denote the resulting fixed fitted function and define its sample-specific target risk and its algorithm-induced expected risk by

$$\mathcal{R}_{\text{fit}}(c) = \mathbb{E}_R \left[\{\hat{t}_c(X) - \tau(X)\}^2 \right], \quad (34)$$

$$\mathfrak{R}(c) = \mathbb{E}_{\text{fit}} \{\mathcal{R}_{\text{fit}}(c)\}. \quad (35)$$

Here $\mathbb{E}_{\text{fit}}$ averages over the nuisance and candidate-fitting samples and all prespecified algorithmic randomness used before validation and final evaluation. When the minimum is attained, define the sample-specific final learner-risk oracle

$$c_{\text{risk}}^*(\text{fit}) \in \arg \min_{c \in \mathfrak{C}} \mathcal{R}_{\text{fit}}(c).$$

This oracle chooses among fitted learning algorithms. It is distinct from both λ_Q^* and $\omega_{\text{LV}}^*(t; \lambda)$.

To make the fallback comparison exact, let

$$\mathfrak{C}_R = \{(0, 0, p, \rho) : (p, \rho) \in \mathcal{H}_R\} \subseteq \mathfrak{C} \quad (36)$$

contain the identical RCT-only learners over the prespecified sieve and penalty grid $\mathcal{H}_R$. Identical means that the RCT observations, folds, RCT nuisance fits, basis, penalty scaling, and optimization code agree with the corresponding fusion candidates; only the two OBS fusion channels are set to zero.

Proposition 10 (Candidate-class oracle dominance). For every realized nuisance and candidate-fitting sample for which the displayed infima are finite,

$$\inf_{c \in \mathfrak{C}} \mathcal{R}_{\text{fit}}(c) \leq \inf_{c \in \mathfrak{C}_R} \mathcal{R}_{\text{fit}}(c). \quad (37)$$

Likewise,

$$\inf_{c \in \mathfrak{C}} \mathfrak{R}(c) \leq \inf_{c \in \mathfrak{C}_R} \mathfrak{R}(c). \quad (38)$$

If the infima are attained, they define the sample-specific and expected-training final CATE-risk oracles. Either inequality is strict only when at least one fusion candidate has risk strictly below the optimally tuned RCT-only class under the corresponding comparison.

Proposition 10 is an infeasible oracle statement because its risks use the unknown τ . It does not say that a data-adaptive selector dominates the RCT-only learner on every finite sample.

Common-reference honest validation. Hold the nuisance-sample and candidate-fitting objects fixed and construct one shared RCT-valid pseudo-outcome $Z_{\lambda_{\text{ref}}}$ on the untouched RCT validation

portion of D_R^{tune} using a fixed reference coefficient. Let $\mathbb{E}_{\text{val}}$ denote expectation only over the prespecified RCT and OBS validation portions of the tuning samples, and let $\mathbb{P}_{R,\text{val}}^{\text{tune}}$ and $\mathbb{P}_{O,\text{val}}^{\text{tune}}$ denote their empirical averages. For candidate c , set

$$G_c(X) = \{g(X) - \hat{t}_c(X)\}^2. \quad (39)$$

Let $r^\dagger = 1$ under the common covariate marginal, $r^\dagger = r$ under exact covariate transport, and $r^\dagger = \hat{r}$ in the estimated-ratio regime. Define the shared-reference validation score

$$\begin{aligned} \hat{\mathcal{R}}_c^{\text{ref}} &= \mathbb{P}_{R,\text{val}}^{\text{tune}} \left[\{Z_{\lambda_{\text{ref}}} - \hat{t}_c(X)\}^2 \right] \\ &+ \mathbb{P}_{O,\text{val}}^{\text{tune}} \{r^\dagger(X)G_c(X)\} - \mathbb{P}_{R,\text{val}}^{\text{tune}} G_c(X). \end{aligned} \quad (40)$$

All candidates must use the same reference pseudo-outcome, g , ratio rule, and validation observations. Under the common marginal or exact r , known RCT propensity, and Assumption 4,

$$\mathbb{E}_{\text{val}}\{\hat{\mathcal{R}}_c^{\text{ref}}\} = Q(\lambda_{\text{ref}}) + \mathcal{R}_{\text{fit}}(c). \quad (41)$$

Thus candidate comparisons remove the same additive reference noise. The OBS correction is not required for validity of this identity, but it can reduce the sampling variability of validation when the prediction is useful.

For the approximate regime, define the fixed-candidate reference bias

$$b_{\text{ref}}(X) = \mathbb{E}_R\{Z_{\lambda_{\text{ref}}} - \tau(X) \mid X\}, \quad (42)$$

the ratio and nuisance drifts

$$\delta_{r,c} = \mathbb{E}_O[\{r^\dagger(X) - r(X)\}G_c(X)], \quad (43)$$

$$\delta_{\text{nuis},c} = 2\mathbb{E}_R[b_{\text{ref}}(X)\{\tau(X) - \hat{t}_c(X)\}], \quad (44)$$

and their uniform magnitudes

$$\Delta_r = \sup_{c \in \mathfrak{C}} |\delta_{r,c}|, \quad \Delta_{\text{nuis}} = \sup_{c \in \mathfrak{C}} |\delta_{\text{nuis},c}|. \quad (45)$$

Known trial propensity and an honest reference nuisance give $b_{\text{ref}} = 0$ for every outcome-regression fit. Exact transport gives $\Delta_r = 0$. Estimated ratios or estimated propensities must instead be controlled through the displayed drifts or a stronger problem-specific expansion. Adaptive reuse of validation observations violates the held-out product law and measurability used here; it is outside Proposition 11 unless a separate selective-inference or stability theorem is supplied.

Proposition 11 (Validation-selector oracle inequality). Let

$$\hat{c} \in \arg \min_{c \in \mathfrak{C}} \hat{\mathcal{R}}_c^{\text{ref}} \quad (46)$$

and define the validation fluctuation

$$\varepsilon_{\text{val}} = \sup_{c \in \mathfrak{C}} \left| \hat{\mathcal{R}}_c^{\text{ref}} - \mathbb{E}_{\text{val}}\{\hat{\mathcal{R}}_c^{\text{ref}}\} \right|. \quad (47)$$

Then

$$\mathcal{R}_{\text{fit}}(\hat{c}) \leq \inf_{c \in \mathfrak{C}} \mathcal{R}_{\text{fit}}(c) + 2\varepsilon_{\text{val}} + 2\Delta_r + 2\Delta_{\text{nuis}} \quad (48)$$

and therefore the same upper bound holds with the infimum over $\mathfrak{C}$ replaced by the larger infimum over $\mathfrak{C}_R$.

If $\varepsilon_{\text{val}} + \Delta_r + \Delta_{\text{nuis}} = o_p(1)$, Proposition 11 gives asymptotic weak noninferiority relative to the identical optimally tuned RCT-only class. Passing from this fixed-candidate statement to expected-training risk additionally requires uniform integrability. Strict asymptotic improvement requires a positive oracle gap that dominates the validation and drift remainders. No part of this argument gives universal finite-sample or realized-sample dominance.

Why λ_Q^* need not optimize sieve risk. Consider an unpenalized linear sieve fitted on RCT pseudo-outcomes with $\omega = 0$, basis b_p , population Gram matrix $G_p = \mathbb{E}_R\{b_p(X)b_p(X)^\top\}$, projection t_p of τ , and leverage $h_p(X) = b_p(X)^\top G_p^{-1} b_p(X)$. Under the usual fixed-sieve least-squares expansion, the λ -dependent part of the leading integrated estimation variance is

$$\frac{1}{n} \mathbb{E}_R [h_p(X) \text{Var}_R(Z_\lambda \mid X)]. \quad (49)$$

The oracle λ_Q^* minimizes the same expression without $h_p(X)$. The two coefficients coincide when the leverage is constant, or under another condition that makes the two weighted quadratics share a minimizer, but they need not coincide in general. Penalties, ω , and data-adaptive basis selection create further differences between the integrated-noise oracle and the final learner-risk oracle.

For $\omega \neq 1$, define $\tilde{Z}_{\lambda,\omega} = (Z_\lambda - \omega g)/(1 - \omega)$.

Proposition 12 (Exact square completion). For every $\omega \neq 1$,

$$\{Z_\lambda - t(X)\}^2 - \omega\{g(X) - t(X)\}^2 = (1 - \omega)\{t(X) - \tilde{Z}_{\lambda,\omega}\}^2 - \frac{\omega}{1 - \omega}\{Z_\lambda - g(X)\}^2.$$

Consequently,

$$\hat{L}_{\text{DR}}(t; \lambda, \omega) = (1 - \omega)\mathbb{P}_{R,n}\{t(X) - \tilde{Z}_{\lambda,\omega}\}^2 \quad (50)$$

$$+ \omega\mathbb{P}_{O,N}\{r(X)(g(X) - t(X))^2\} + \left[-\frac{\omega}{1 - \omega}\mathbb{P}_{R,n}\{Z_\lambda - g(X)\}^2\right]. \quad (51)$$

The bracketed term is constant in t .

For $0 \leq \omega < 1$, the two t -dependent terms form a nonnegative mixture of squared losses. At $\omega = 1$, the transformed outcome is undefined and the original objective must be used. For $0 \leq \omega \leq 1$, the original empirical quadratic is positive semidefinite in unrestricted fitted values and in the coefficients of a linear sieve because its Hessian is a nonnegative weighted sum of Gram matrices. Outside this interval, or for nonlinear parameterizations, convexity is not guaranteed. Under a penalty or parameter constraint, rescaling the completed loss changes the relative penalty scale, so the original quadratic and its stated penalty must define the estimator.

4.2. Implementable F-Learner and prospective R-loss extension

We use *F-Learner* as a provisional working label only for CATE estimators that minimize Equation (28), which is built from the RCT-valid AIPW pseudo-outcome Z_λ . The label does not claim name novelty and does not denote a generic unification of doubly robust and R losses. Let $\mathbb{P}_R^{\text{fit}}$ and $\mathbb{P}_O^{\text{fit}}$ denote empirical averages over the prespecified candidate-fitting portions of D_R^{tune} and D_O^{tune} , respectively.

For a linear sieve $t_\beta(x) = b(x)^\top \beta$, with $b(x) \in \mathbb{R}^p$, add the penalty $\rho \beta^\top P \beta$, where $\rho \geq 0$ and $P = P^\top \succeq 0$ belongs to $\mathbb{R}^{p \times p}$. Define

$$G_R = \mathbb{P}_R^{\text{fit}}\{b(X)b(X)^\top\}, \quad G_{O,r} = \mathbb{P}_O^{\text{fit}}\{r(X)b(X)b(X)^\top\}, \quad (52)$$

$$h_\lambda = \mathbb{P}_R^{\text{fit}}\{b(X)Z_\lambda\}, \quad d_g = \mathbb{P}_O^{\text{fit}}\{r(X)b(X)g(X)\} - \mathbb{P}_R^{\text{fit}}\{b(X)g(X)\}. \quad (53)$$

Whenever the matrix is invertible, the penalized F-Learner solution is

$$\hat{\beta}_{\lambda,\omega} = \{(1-\omega)G_R + \omega G_{O,r} + \rho P\}^{-1} \{h_\lambda + \omega d_g\}. \quad (54)$$

Because h_λ is affine in λ , a fixed- ω candidate path can be computed efficiently. A frozen neural basis replaces $b(x)$ by a representation learned without the candidate-training or validation observations and inherits Equation (54) conditional on that representation. End-to-end neural fitting is nonconvex, and the present linear-sieve guarantees do not cover it.

Out-of-fold nuisance and candidate construction. As an alternative within the nuisance and tuning stages, partition the RCT and OBS indices into prespecified folds. For each held-out fold k , fit $\hat{\mu}_R^{(-k)}$ and $\hat{\mu}_O^{(-k)}$ without that fold, use the same

$$g^{(-k)}(x) = \hat{\mu}_{O,1}^{(-k)}(x) - \hat{\mu}_{O,0}^{(-k)}(x)$$

on held-out RCT and OBS covariates, and construct $Z_{0,i}^{(-k)}$, $Z_{1,i}^{(-k)}$, and $Z_{\lambda,i}^{(-k)} = Z_{0,i}^{(-k)} + \lambda\{Z_{1,i}^{(-k)} - Z_{0,i}^{(-k)}\}$ only on the held-out RCT fold. This prevents an observation's own outcome from entering its nuisance prediction, but estimates from different folds remain dependent.

For each $c \in \mathfrak{C}$, candidate fitting, basis construction, penalty selection, and tuning-dependent preprocessing must exclude the validation portion of D_S^{tune} and all of D_S^{eval} . The common-reference validation score then ranks candidates already fitted using the nuisance samples and the fitting portion of the tuning samples. Reusing validation outcomes to fit a candidate or its preprocessing violates the held-out product law and lies outside Proposition 11.

A prospective R-loss extension can be implemented as an empirical candidate, but it requires a separate theorem before it belongs to the F-Learner class. Let m be a generic fixed, cross-fitted marginal-outcome regression trained independently of the observations on which the following loss is evaluated. One candidate objective is

$$\hat{L}_R(t; \omega) = \mathbb{P}_{R,n}[\{Y - m(X)\} - \{A - e(X)\}t(X)]^2 \quad (55)$$

$$+ \omega \left[\mathbb{P}_{O,N}\{r(X)(g(X) - t(X))^2\} - \mathbb{P}_{R,n}\{(g(X) - t(X))^2\} \right], \quad (56)$$

Randomized treatment implies

$$\mathbb{E}_R[(A - e(X))\{Y - m(X)\} \mid X] = e(X)\{1 - e(X)\}\tau(X) \quad (57)$$

for every fixed m . This identity motivates the residualized objective. No λ channel has yet been derived for this R-loss candidate. A complete result must still derive the precise target under penalties and restricted learners, a principled role for λ , an honest validation score, and estimated- nuisance rates. We therefore present Equation (56) as a planned extension rather than a proved contribution.

Under covariate shift, an exact ratio can be recovered from domain-classification odds. If $S = 1$ denotes RCT membership, $\pi = P(S = 1)$, and $s(x) = P(S = 1 \mid X = x)$, then

$$r(x) = \frac{1 - \pi}{\pi} \frac{s(x)}{1 - s(x)}. \quad (58)$$

An estimated ratio $\hat{r}$ replaces exact target preservation with an explicit drift. Clipping changes the exact transported target and must be reported.

A loss-targeted alternative chooses a nonnegative, out-of-sample-evaluable weight function w to balance

$$\mathcal{G}(\mathcal{T}; g) = \{(g - t)^2 : t \in \mathcal{T}\}. \quad (59)$$

If $\mathbb{E}_O\{w(g - t)^2\} = \mathbb{E}_R\{(g - t)^2\}$ for every $t \in \mathcal{T}$, Equation (30) remains exact with w in place of r . For a linear sieve, balancing bg and $bb^\top$ preserves the minimizer; balancing g^2 is additionally required only to preserve the absolute loss value. Exact balancing on the same observations used to fit t makes the correction zero or constant in the sieve coefficient. Weight learning and candidate fitting must therefore use independent samples or appropriate outer folds.

For approximate weights, define

$$\Delta_{\mathcal{T}}(w) = \sup_{t \in \mathcal{T}} \left| \mathbb{E}_O\{w(X)(g(X) - t(X))^2\} - \mathbb{E}_R\{(g(X) - t(X))^2\} \right|. \quad (60)$$

The uniform population-loss drift is bounded by $|\omega|\Delta_{\mathcal{T}}(w)$, and the excess target risk of the weighted population minimizer is at most $2|\omega|\Delta_{\mathcal{T}}(w)$ when both minimizers exist.

Current scope and open theory. The proved CATE statements cover the AIPW-pseudo-outcome squared loss with fixed honest nuisances and either a common covariate marginal or exact transport. The linear-sieve solution and frozen-feature extension are conditional results. Estimated-ratio rates, adaptive loss-targeted balancing, end-to-end neural optimization, the R-loss extension, and cross-fitted inference remain open theory. Honest splitting prevents direct reuse but does not by itself close these gaps.

A reproducible analysis should report the candidate class $\mathfrak{C}$, RCT-only subclass $\mathfrak{C}_R$, folds, basis dimension, penalty and scaling, ratio method and overlap diagnostics, denominator fallbacks, selected candidate, and validation score of the identical RCT-only comparator. Values of ω outside $[0, 1]$ require an explicit curvature check of the original empirical objective.

5. Experiments

5.1. Completed ATE benchmarks

The completed experiments evaluate ATE estimation only. All nested benchmarks use 20 Monte Carlo repetitions per cell, an RCT-only AIPW baseline, a λ -only outcome-model fusion estimator, an ω -only PPI correction estimator, and the joint estimator. The code retains two stale internal names: `haipw_only` denotes the λ -only estimator and `ppi_power_tuned_only` denotes the ω -only estimator. The first label does not identify the method with the separate HAIPW literature.

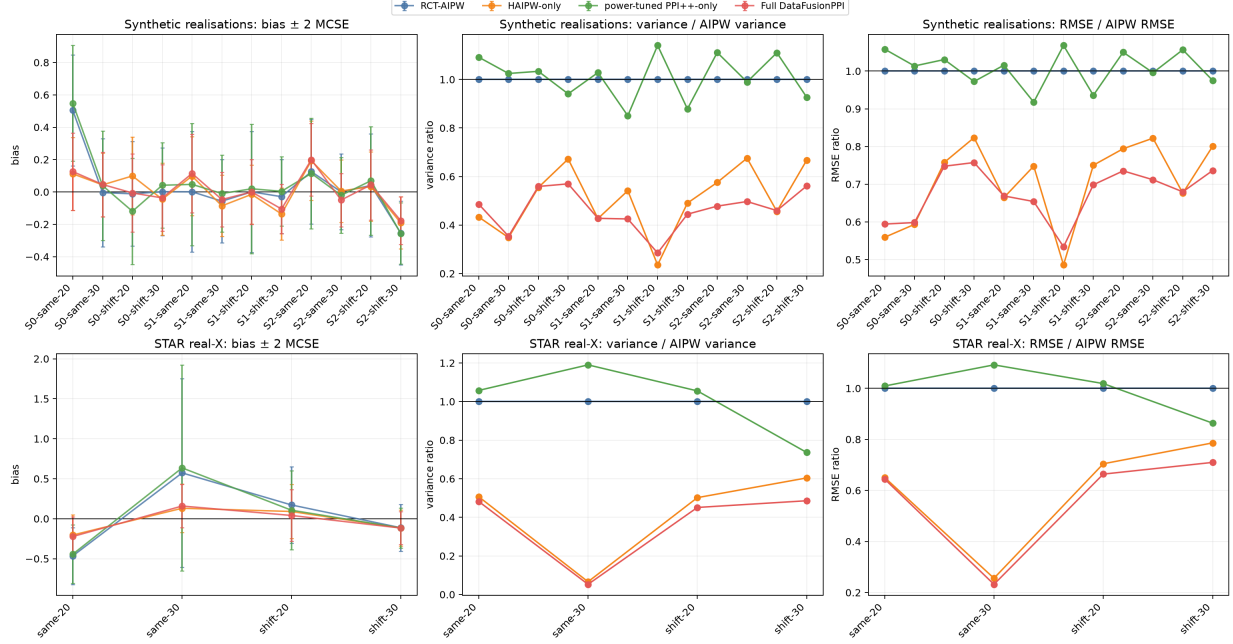


Figure 1: Fixed-OBS-budget ATE benchmark. The composite artifact reports multiple method and regime panels side by side for $n_R \in \{20, 30\}$ and $N_O = 5000$.

The fixed-OBS-budget benchmark contains 16 design cells: three arbitrary nonlinear SCMs and one STAR real- X construction, each under a common covariate marginal and covariate shift and $n_R \in \{20, 30\}$, with $N_O = 5000$. The joint estimator has the smallest RMSE in 11 cells, and the λ -only estimator is best in the remaining five. The joint estimator has lower RMSE than AIPW in all 16 cells, with mean RMSE ratio 0.648 and mean empirical-variance ratio 0.439 relative to AIPW. The λ -only estimator also beats AIPW in all 16 cells, with mean RMSE ratio 0.679. The ω -only estimator is not best in any cell and has mean RMSE ratio 1.005.

The proportional-budget benchmark contains 32 cells with $n_R \in \{20, 30, 50, 100\}$ and $N_O = 100n_R$. The joint estimator has the smallest RMSE in 18 cells, the λ -only estimator in 10, the ω -only estimator in one, and AIPW in three. The joint estimator has lower RMSE than AIPW in 27 of 32 cells, with mean RMSE ratio 0.760 and mean empirical-variance ratio 0.625. The λ -only estimator beats AIPW in 28 cells and has mean RMSE ratio 0.769. The ω -only estimator beats AIPW in 10 cells and has mean RMSE ratio 1.020. These counts show that joint tuning is often useful but is not uniformly best after coefficient estimation.

The STAR benchmark uses only real kindergarten covariates from Project STAR. Treatment assign-

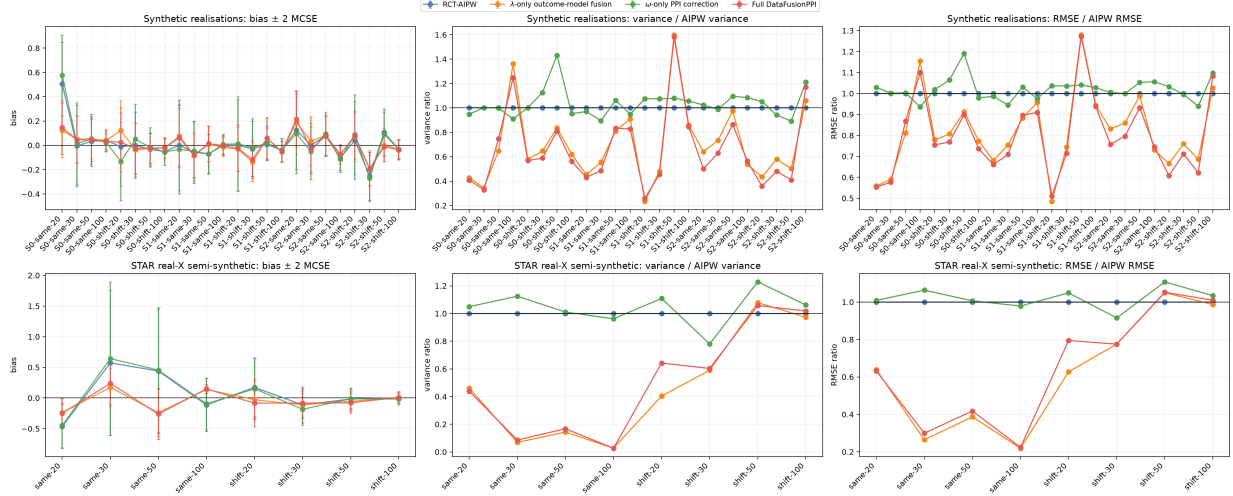


Figure 2: Proportional-budget ATE benchmark. The composite artifact reports multiple method and regime panels side by side for $n_R \in \{20, 30, 50, 100\}$ and $N_O = 100n_R$.

ments, latent confounding, potential outcomes, and effect heterogeneity are generated synthetically. It is therefore a real- X , semi-synthetic benchmark rather than an analysis of STAR’s original intervention or outcomes. The planned WHI CT+OS analysis has not been run because participant-level access, a semantic crosswalk, and a survival or censoring extension are still required.

5.2. Planned CATE evaluation and interpretation

CATE experiments are planned. The primary design will compare the proved AIPW-pseudo-outcome F-Learner with the prospective R-loss extension using additive cubic B-splines, frozen neural bases, and linear sieves under a common covariate marginal and covariate shift. Each study will compare trial-only, λ -only, ω -only, and joint tuning using held-out RCT-population CATE risk. Classifier-only, balancing-only, hybrid, and oracle transport weights will be separated. No CATE result is claimed from Figures 1 and 2.

The empirical conclusion is deliberately limited. The two-channel population oracle weakly improves the ATE variance objective when the trial-only coefficient is feasible, but estimated coefficients do not inherit a universal no-harm property. The present 20-repetition studies suggest that the λ channel is the more stable source of gain and that joint tuning can add value in some designs. More Monte Carlo repetitions, completed CATE experiments, additional real- X supports, and an authorized real-outcome analysis are required before making a stronger empirical claim.

5.3. Scope, failure modes, and conclusion

The framework uses observational information through prediction rather than by declaring the observational treatment contrast causal. Trial randomization validates the pseudo-outcome for every fixed outcome regression. A common covariate marginal or an exact density ratio centers the ATE control variate and the prediction-powered CATE loss, which creates separate λ - and ω -channels while retaining RCT-population targets.

The first boundary is covariate transport. Without weighting under $P_R^X \neq P_O^X$, the ATE correction has drift $\omega\{P_O g - P_R g\}$, and the unweighted CATE loss targets the wrong covariate average. Exact $r = dP_R^X/dP_O^X$ removes both discrepancies. An estimated ratio instead contributes the explicit ATE drift $\omega P_O\{(\hat{r} - r)g\}$ and the CATE validation drift in Equation (45). Evaluation-only variance is first-order adequate only under the stated negligible-drift conditions; otherwise ratio-training influence and covariance terms are required.

The second boundary is trial propensity overlap. Proposition 1 uses the known, strictly positive randomized-trial propensity. Extreme inverse-propensity factors can make the required moments unstable even when the conditional-mean identity remains algebraically valid, and an incorrect propensity removes the exact arbitrary- q guarantee.

The third boundary is weak observational prediction. The unrestricted ω -gain depends on how strongly $g(X)$ predicts $\tau(X)$, with attenuation from the finite n/N ratio. The λ -channel gives little gain when $\Delta = Z_1 - Z_0$ is uninformative or nearly degenerate. Population oracles diagnose these limits but do not make their empirical coefficient estimates stable.

The fourth boundary is adaptive reuse. Fitting nuisances, tuning candidates, and evaluating the selected candidate on the same outcomes invalidates the conditioning product law. Independent nuisance, tuning, and evaluation samples prevent direct leakage; they may be supplied separately or constructed by splitting. Optional nested outer cross-fitting preserves foldwise and aggregate unbiasedness under exact transport and fold honesty, but overlapping training complements invalidate a naive sum of fold variances. Cross-fitted inference needs an additional stability or influence-function theorem.

The fifth boundary is learner restriction and curvature. Target preservation does not make a finite-sample restricted or highly penalized projection accurate. The square-completed representation is a nonnegative mixture only for $0 \leq \omega < 1$, and positive semidefiniteness at $\omega = 1$ must be read from the original quadratic. Coefficients outside $[0, 1]$, nonlinear parameterizations, and penalty rescaling require direct analysis of the original objective.

The sixth boundary is tail behavior. ATE variance identities require square-integrable scores; the Wald theorem additionally requires conditional Lindeberg control, a sufficient $(2 + \delta)$ -moment condition, and a nondegenerate variance limit. Fixed-loss variance tuning imposes fourth-moment-type requirements. Heavy tails can therefore destabilize oracle formulas even when their algebraic denominators are nonzero.

Within these boundaries, the randomized trial fixes the causal target, the OBS outcome model can reduce within- X pseudo-outcome noise through λ , and the OBS covariate sample can reduce predictable between- X variation through ω . The exact results are conditional identities and population-oracle comparisons. Empirical work must retain the identical trial-only fallback, report selection and denominator instability, and avoid interpreting them as a universal finite-sample no-harm guarantee.

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A. Simulation Details

All completed simulation artifacts were generated with base seed 190602. The primary nested benchmarks use 20 outer Monte Carlo repetitions per design cell and compare four estimators: RCT-only AIPW, λ -only outcome-model fusion, ω -only PPI correction, and joint (λ, ω) fusion. The implementation retains the historical internal codes `haipw_only` and `ppi_power_tuned_only` for the second and third estimators, respectively. These codes are not the user-facing method names, and the first does not refer to the separate HAIPW paper.

The arbitrary simulation family uses three nonlinear structural causal models. In every model, a domain indicator S distinguishes the observational sample ($S = 0$) from the randomized sample ($S = 1$); covariates depend on latent variables, observational treatment depends on both measured covariates and an unmeasured confounder, trial treatment follows the known randomized propensity, and outcomes may depend nonlinearly on covariates, treatment, and the latent confounder. The common-marginal regime uses $P_R^X = P_O^X$. The shifted regime changes the covariate distribution while preserving overlap and estimates the density ratio from a domain classifier. The same estimators and evaluation logic are used in both regimes.

The fixed-OBS-budget experiment has 16 cells: four benchmark families, two covariate regimes, and $n_R \in \{20, 30\}$, with $N_O = 5000$. The proportional experiment has 32 cells: four benchmark families, two regimes, and $n_R \in \{20, 30, 50, 100\}$, with the exact mapping $N_O = 100n_R \in \{2000, 3000, 5000, 10000\}$. Coefficients are estimated by the nested honest procedure; no oracle coefficient or oracle density ratio is used in the reported method comparison. The replication-level CSV files, summaries, and composite plots in `materials/` are the canonical numerical artifacts.

The fourth benchmark family reuses the empirical covariate support from the Tennessee Student/Teacher Achievement Ratio (STAR) experiment. Treatment, latent confounding, outcomes, and treatment-effect heterogeneity are generated by the benchmark code. Consequently this is a real- X , semi-synthetic design, not an evaluation of the original STAR outcomes or intervention effect.

Figure 3 records the earlier two-method pilot. It is retained for provenance but is not the basis for the four-method conclusions in Section 5.

No participant-level WHI CT+OS analysis is included. Such an analysis requires authorized data access, a verified trial-observational semantic crosswalk, a prespecified estimand, and methods appropriate for censoring if the selected outcome is time-to-event. No completed CATE experiment or real-outcome result is represented by the present artifacts.

Illustrative random-SCM debugging pilot (20 outer refits per cell)
per stage: RCT n / OBS 5000; total generated per replicate: $3n / 15000$

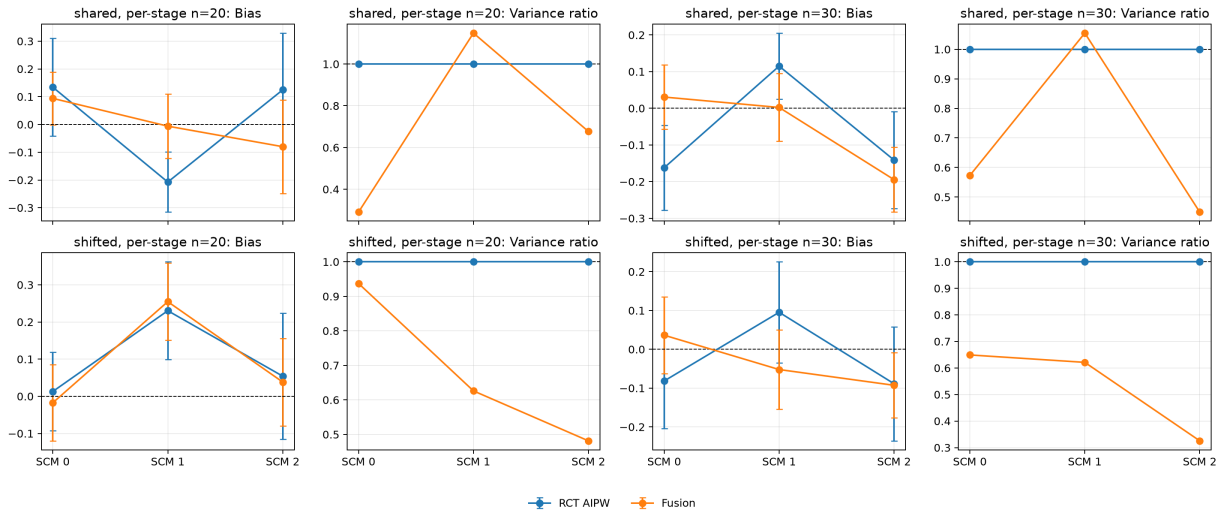


Figure 3: Earlier SSEM ATE pilot. This artifact predates the nested four-method comparison and should be interpreted only as a design and implementation check.

B. Proof Details

Unless stated otherwise, all proofs use Assumption 4: every nuisance, ratio, candidate function, and coefficient is fixed before the independent RCT and OBS evaluation observations are used.

B.1. Conditional validity of the trial pseudo-outcome

Fix $q = (q_0, q_1)$ without using the evaluation samples and condition on X . Under trial randomization, consistency, and honest sample separation,

$$\mathbb{E}_R \left[\frac{A}{e(X)} \{Y - q_1(X)\} \mid X \right] = \mathbb{E}_R \{Y(1) - q_1(X) \mid X\}, \quad (61)$$

$$\mathbb{E}_R \left[\frac{1 - A}{1 - e(X)} \{Y - q_0(X)\} \mid X \right] = \mathbb{E}_R \{Y(0) - q_0(X) \mid X\}. \quad (62)$$

Substituting these two equalities into the definition of φ cancels the two fitted regressions and yields $\mathbb{E}_R \{\varphi(V; q) \mid X\} = \tau(X)$. Applying the identity to $q = \hat{\mu}_R$, $q = \hat{\mu}_O$, and their affine blend gives

$$\mathbb{E}_R(Z_\lambda \mid X) = \tau(X), \quad \mathbb{E}_R(\Delta \mid X) = 0. \quad (63)$$

B.2. Proof of Corollary 1

Apply Equation (7) with the nuisance-sample fitted pair $q = \hat{\mu}_R$, average over D_R^{eval} , and use iterated expectation. This gives

$$\mathbb{E}\{\hat{\theta}_{\text{AIPW}, R}\} = \mathbb{E}_R\{\tau(X)\} = \theta.$$

B.3. Proof of Theorem 1

Using $\mathbb{E}_R(Z_\lambda) = \theta$ and exact transport,

$$\mathbb{E}_O\{r(X)g(X)\} = \mathbb{E}_R\{g(X)\}, \quad (64)$$

the evaluation-sample mean of Equation (9) is θ . Rewrite the centered estimator as

$$\hat{\theta}_r(\lambda, \omega) - \theta = (\mathbb{P}_{R,n} - P_R)\{Z_\lambda - \omega g\} + (\mathbb{P}_{O,N} - P_O)\{\omega r g\}. \quad (65)$$

The two summands are independent because D_R^{eval} and D_O^{eval} are independent. The variance of an empirical mean of m independent observations is the population variance divided by m , which proves Equation (11). Unbiased sample variances on the two independent evaluation blocks therefore give the displayed fixed-selected-coefficient variance estimator in Section 3. Coverage of its Wald interval additionally uses the stated evaluation-sample Lindeberg and moment assumptions; variance unbiasedness alone is not a central limit theorem.

Because $g(X)$ is fixed given X ,

$$\mathbb{E}_R\{Z_\lambda - \omega g(X) \mid X\} = \tau(X) - \omega g(X), \quad \text{Var}_R\{Z_\lambda - \omega g(X) \mid X\} = \text{Var}_R(Z_\lambda \mid X).$$

Applying the law of total variance proves Equation (12).

B.4. Proof of Proposition 2

Expand the first variance in Equation (11) using $Z_\lambda = Z_0 + \lambda\Delta$. After collecting terms and multiplying by n , the coefficient-dependent part is

$$2C\lambda - 2D\omega + A\lambda^2 - 2E\lambda\omega + B\omega^2, \quad (66)$$

which is $\mathcal{J}_{\text{ATE}}(\beta)$. This proves Equation (16). The representation

$$\begin{pmatrix} u & v \end{pmatrix} \mathbf{H}_{\text{ATE}} \begin{pmatrix} u \\ v \end{pmatrix} = \text{Var}_R(u\Delta - vg) + \frac{n}{N}v^2 \text{Var}_O(rg) \quad (67)$$

proves positive semidefiniteness.

B.5. Proof of Proposition 3

Equation (7) applied to Z_0 and Z_1 gives $\mathbb{E}_R(Z_a | X) = \tau(X)$ for $a \in \{0, 1\}$. Subtracting the two identities gives $\mathbb{E}_R(\Delta | X) = 0$. Because g is a fixed function of X , iterated expectation yields

$$\text{Cov}_R(\Delta, g) = \mathbb{E}_R[\{g(X) - \mathbb{E}_R g(X)\} \mathbb{E}_R(\Delta | X)] = 0.$$

The same argument with Z_0 gives $\text{Cov}_R(Z_0, g) = \text{Cov}_R\{\tau(X), g(X)\}$.

B.6. Proof of Proposition 4

Equation (16) is the positive affine transformation

$$\text{Var}\{\hat{\theta}_r(\beta)\} = \frac{1}{n} \text{Var}_R(Z_0) + \frac{1}{n} \mathcal{J}_{\text{ATE}}(\beta),$$

where $n > 0$. Adding a constant and multiplying by a positive scalar preserve the complete argmin set over every feasible set.

B.7. Proof of Theorem 2

If $AB - E^2 > 0$, $\mathbf{H}_{\text{ATE}}$ is positive definite. The first-order condition for $\mathcal{J}_{\text{ATE}}$ is $\mathbf{H}_{\text{ATE}}\beta = -\ell$, which has the unique solution

$$\beta_{\text{joint}}^* = -\mathbf{H}_{\text{ATE}}^{-1}\ell = \begin{pmatrix} (ED - BC)/(AB - E^2) \\ (AD - EC)/(AB - E^2) \end{pmatrix}.$$

For $\rho > 0$, the ridge objective has gradient

$$\nabla_\beta \left\{ \mathcal{J}_{\text{ATE}}(\beta) + \rho \|\beta\|_2^2 \right\} = 2\ell + 2(\mathbf{H}_{\text{ATE}} + \rho I_2)\beta.$$

Positive semidefiniteness of $\mathbf{H}_{\text{ATE}}$ makes $\mathbf{H}_{\text{ATE}} + \rho I_2$ positive definite. Setting the gradient to zero proves Equation (20) and uniqueness.

B.8. Singular and constrained extensions

When $\mathbf{H}_{\text{ATE}}$ is singular, nonnegativity of the variance quadratic implies $\ell \in \text{Range}(\mathbf{H}_{\text{ATE}})$; otherwise a null-space direction would make the expression arbitrarily negative. The complete minimizer set is therefore $-\mathbf{H}_{\text{ATE}}^+ \ell + \text{Null}(\mathbf{H}_{\text{ATE}})$. Continuity of the quadratic proves attainment on every prespecified compact feasible set.

For completeness, suppose $\mathbf{H}_{\text{ATE}}$ is positive definite and write

$$\mathcal{J}_{\text{ATE}}(\beta) = 2\ell^\top \beta + \beta^\top \mathbf{H}_{\text{ATE}} \beta, \quad \beta_{\text{joint}}^* = -\mathbf{H}_{\text{ATE}}^{-1} \ell.$$

Completing the square gives

$$\mathcal{J}_{\text{ATE}}(\beta) - \mathcal{J}_{\text{ATE}}(\beta_{\text{joint}}^*) = (\beta - \beta_{\text{joint}}^*)^\top \mathbf{H}_{\text{ATE}} (\beta - \beta_{\text{joint}}^*).$$

The norm induced by a positive-definite matrix is equivalent to the Euclidean norm. Hence this distance is coercive and strictly convex. On every nonempty closed convex feasible set, it therefore attains a unique minimum, namely the $\mathbf{H}_{\text{ATE}}$ -metric projection of β_{joint}^* .

Finally, because g is X -measurable,

$$E = \mathbb{E}_R[\{g(X) - P_R g\} \mathbb{E}_R(\Delta \mid X)] = 0, \quad (68)$$

$$D = \text{Cov}_R\{\mathbb{E}_R(Z_0 \mid X), g(X)\} = \text{Cov}_R\{\tau(X), g(X)\}. \quad (69)$$

If $A = \text{Var}_R(\Delta) = 0$, then Δ is constant almost surely, and $\mathbb{E}_R(\Delta \mid X) = 0$ forces that constant to be zero. If

$$B = \text{Var}_R(g) + \frac{n}{N} \text{Var}_O(rg) = 0,$$

then g is constant P_R -almost surely and rg is constant P_O -almost surely. Exact transport gives $P_O(rg) = P_R(g)$, so the two constants agree and $\mathbb{P}_{O,N}(rg) - \mathbb{P}_{R,n}(g) = 0$ for every evaluation sample. Finally, when $E = 0$ and $A, B > 0$, completing the two scalar squares gives

$$\mathcal{J}_{\text{ATE}}(\beta) = A \left(\lambda + \frac{C}{A} \right)^2 + B \left(\omega - \frac{D}{B} \right)^2 - \frac{C^2}{A} - \frac{D^2}{B}.$$

Therefore minimization over a rectangular feasible set is exactly coordinatewise clipping of $-C/A$ and D/B . This argument does not apply to coupled constraints or to a nonzero interaction E . Substitution yields Equation (21). Since the population objective is minimized over a set containing $(0, 0)$, its oracle value cannot exceed the trial-only value. This comparison concerns the population objective, not coefficients estimated from a finite tuning sample.

B.9. Proof of Corollary 3

Write the evaluation-sample variance as

$$V(\beta) = \mathcal{V}_0 + \frac{1}{n} \{2\ell^\top \beta + \beta^\top \mathbf{H}_{\text{ATE}} \beta\}, \quad \mathcal{V}_0 = \frac{1}{n} \text{Var}_R(Z_0).$$

When $AB - E^2 > 0$, the matrix $\mathbf{H}_{\text{ATE}}$ is positive definite and the unique minimizer is $\beta_{\text{joint}}^* = -\mathbf{H}_{\text{ATE}}^{-1} \ell$. Substitution gives

$$V(\beta_{\text{joint}}^*) = \mathcal{V}_0 - \frac{1}{n} \ell^\top \mathbf{H}_{\text{ATE}}^{-1} \ell.$$

For $\mathbf{H}_{\text{ATE}} = \begin{pmatrix} A & -E \\ -E & B \end{pmatrix}$,

$$\begin{aligned} \mathbf{H}_{\text{ATE}}^{-1} &= \frac{1}{AB - E^2} \begin{pmatrix} B & E \\ E & A \end{pmatrix}, \\ \ell^\top \mathbf{H}_{\text{ATE}}^{-1} \ell &= \frac{BC^2 - 2ECD + AD^2}{AB - E^2}. \end{aligned}$$

Positive definiteness makes this quadratic form nonnegative and makes it zero exactly when $\ell = 0$. Setting $E = 0$ yields $C^2/A + D^2/B$.

Singular variance reduction and compact constraints. If $\mathbf{H}_{\text{ATE}}$ is singular, boundedness below of the variance objective implies $\ell \in \text{Range}(\mathbf{H}_{\text{ATE}})$. Evaluating at the minimum-norm solution $-\mathbf{H}_{\text{ATE}}^+ \ell$ gives

$$V_{\min} = \mathcal{V}_0 - \frac{1}{n} \ell^\top \mathbf{H}_{\text{ATE}}^+ \ell.$$

The pseudoinverse is positive definite on $\text{Range}(\mathbf{H}_{\text{ATE}})$, so the reduction is strictly positive exactly when $\ell \neq 0$. Finally, continuity gives attainment on every nonempty compact feasible set $\mathcal{B}_\beta$. If $0 \in \mathcal{B}_\beta$, the constrained minimum is at most $V(0) = \mathcal{V}_0$, and it is strictly smaller exactly when some feasible β makes $2\ell^\top \beta + \beta^\top \mathbf{H}_{\text{ATE}} \beta < 0$.

B.10. Prediction-powered CATE risk identity

Fix t, λ, ω, g, r before using the evaluation samples, and then condition on X . Since $\mathbb{E}_R(Z_\lambda - \tau(X) \mid X) = 0$,

$$\mathbb{E}_R\{Z_\lambda - t(X)\}^2 = \mathbb{E}_R\{Z_\lambda - \tau(X)\}^2 + \mathbb{E}_R\{t(X) - \tau(X)\}^2. \quad (70)$$

Exact transport also gives

$$\mathbb{E}_O\{r(X)(g(X) - t(X))^2\} = \mathbb{E}_R\{g(X) - t(X)\}^2. \quad (71)$$

The correction in Equation (28) therefore has mean zero, which proves Equation (30). In particular, population expected loss is flat in ω . The part depending on λ is

$$Q(\lambda) = Q(0) + 2C\lambda + A\lambda^2, \quad (72)$$

so when $A > 0$ its minimizer is $-C/A$, the same integrated-noise oracle as in the population-orthogonal ATE problem. If $A = 0$, then $\Delta = 0$ almost surely by conditional mean zero, and Q is flat in λ .

After fixing t , λ , g , and r using the nuisance and tuning samples, the empirical loss is the sum of independent empirical means of $U_{t,\lambda} - \omega G_t$ over D_R^{eval} and $\omega r G_t$ over D_O^{eval} . Expanding its variance, with sample-size ratio n/N , as a quadratic in ω and differentiating gives Equation (32). Because t is fixed in this calculation, the formula does not identify the coefficient minimizing the risk of a learner chosen after seeing the same loss.

For the linear sieve, evaluate the empirical objective on the prespecified candidate-fitting portions of the tuning samples, using $\mathbb{P}_R^{\text{fit}}$ and $\mathbb{P}_O^{\text{fit}}$. Expand the result in β , add $\rho\beta^\top P\beta$, and set the gradient to zero. The resulting normal equation is

$$\{(1 - \omega)G_R + \omega G_{O,r} + \rho P\}\beta = h_\lambda + \omega d_g, \quad (73)$$

which proves Equation (54) whenever the matrix is invertible.

B.11. Pointwise noise coefficients and square completion

For Proposition 8, measurability and conditional mean zero give

$$\mathbb{E}_R\{Z_0 + \lambda(X)\Delta \mid X\} = \tau(X) + \lambda(X)\mathbb{E}_R(\Delta \mid X) = \tau(X).$$

At a fixed x , expand the conditional noise as a quadratic in a scalar u :

$$Q_x(u) = Q_x(0) + 2u \text{Cov}_R(Z_0, \Delta \mid X = x) + u^2 \text{Var}_R(\Delta \mid X = x).$$

A positive quadratic coefficient yields the displayed pointwise ratio. If it is zero, $\Delta = 0$ conditionally almost surely and every u -dependent term vanishes.

For Proposition 12, expand the trial contribution as

$$\{Z_\lambda - t\}^2 - \omega\{g - t\}^2 = (1 - \omega)t^2 - 2(Z_\lambda - \omega g)t + Z_\lambda^2 - \omega g^2.$$

Since $(1 - \omega)\tilde{Z}_{\lambda,\omega} = Z_\lambda - \omega g$, completing the square leaves

$$Z_\lambda^2 - \omega g^2 - (1 - \omega)\tilde{Z}_{\lambda,\omega}^2 = -\frac{\omega}{1 - \omega}(Z_\lambda - g)^2.$$

Averaging and adding the transported OBS term proves the proposition. For a linear sieve, the Hessian of the original t -dependent quadratic is a nonnegative weighted sum of G_R and $G_{O,r}$ for $0 \leq \omega \leq 1$, which proves positive semidefiniteness in that range. The claim does not extend automatically to nonlinear parameterizations.

B.12. Loss-targeted balance and approximate drift

The two inequalities in Proposition 10 follow directly from the prespecified inclusion $\mathfrak{C}_R \subseteq \mathfrak{C}$. Strict inclusion alone does not imply a strict risk inequality.

For Proposition 11, fix the candidates fitted using the nuisance samples and the fitting portion of the tuning samples, then take expectation over the disjoint validation portion. Expanding the shared-reference score gives

$$\mathbb{E}_{\text{val}}\{\widehat{\mathcal{R}}_c^{\text{ref}}\} = Q(\lambda_{\text{ref}}) + \mathcal{R}_{\text{fit}}(c) + \delta_{r,c} + \delta_{\text{nuis},c}.$$

Compare the empirical score of $\widehat{c}$ with that of an arbitrary $c \in \mathfrak{C}$. Two validation deviations contribute at most $2\varepsilon_{\text{val}}$, and the difference between two ratio and nuisance drifts contributes at most $2\Delta_r + 2\Delta_{\text{nuis}}$. Taking the infimum gives the stated oracle inequality.

Suppose a nonnegative out-of-sample-evaluable weight w satisfies

$$P_O\{w(g-t)^2\} = P_R\{(g-t)^2\} \quad \text{for every } t \in \mathcal{T}. \quad (74)$$

Replacing r by w then leaves the population correction exactly zero for every candidate, so the target identity remains valid. If $t_\beta = b^\top \beta$, expanding

$$(g - b^\top \beta)^2 = g^2 - 2\beta^\top b g + \beta^\top b b^\top \beta \quad (75)$$

shows that balancing bg and $bb^\top$ preserves the minimizer. Balancing g^2 is needed only to preserve the absolute loss value.

For approximate weights, the difference between the weighted and target population losses is bounded uniformly over $\mathcal{T}$ by $|\omega|\Delta_{\mathcal{T}}(w)$. Let t_w minimize the weighted population loss and t_0 minimize the target loss. Applying the uniform bound at both minimizers gives

$$L_0(t_w) \leq L_w(t_w) + |\omega|\Delta_{\mathcal{T}}(w) \quad (76)$$

$$\leq L_w(t_0) + |\omega|\Delta_{\mathcal{T}}(w) \quad (77)$$

$$\leq L_0(t_0) + 2|\omega|\Delta_{\mathcal{T}}(w), \quad (78)$$

which is the claimed excess-risk bound. If the same finite sample is used both to impose exact sieve balance and to fit β , the balanced linear and quadratic coefficient terms cancel from the empirical correction (and the constant cancels as well if g^2 is balanced). This algebra explains why weight learning and candidate fitting require honest separation or a separate outer-fold argument.

B.13. A two-stratum oracle calculation

Let $P_R(X=0) = P_R(X=1) = 1/2$, $\tau(0) = 0$, $\tau(1) = 2$, and $g = \tau$. With $n = 100$, $N = 900$, and a common covariate marginal,

$$\mathbb{E}_R\{\tau(X)\} = 1, \quad \text{Var}_R\{\tau(X)\} = 1, \quad \omega_{\text{ATE}}^* = \frac{1}{1 + 100/900} = 0.9.$$

The baseline between- X variance is 0.01, whereas the oracle component is

$$\frac{\text{Var}_R\{0.1\tau(X)\}}{100} + \frac{0.9^2 \text{Var}_R\{\tau(X)\}}{900} = 0.0001 + 0.0009 = 0.001.$$

The reduction 0.009 concerns only between- X variation; total variance also contains the within- X pseudo-outcome noise from Equation (12).

B.14. Honest variance estimation and evaluation-sample Wald inference

For Proposition 5, fix the coefficients and scores before using the evaluation observations. Ordinary unbiased sample-variance identities give

$$\mathbb{E}\{\widehat{\text{Var}}_{R,\text{eval}}(\psi_R)\} = \text{Var}_R(\psi_R), \quad \mathbb{E}\{\widehat{\text{Var}}_{O,\text{eval}}(\psi_O)\} = \text{Var}_O(\psi_O).$$

Dividing by n and N , then adding, proves unbiasedness of $\widehat{V}$ by independence of D_R^{eval} and D_O^{eval} .

For Theorem 3, center the two evaluation-score arrays at their population means after fixing the learned objects. Their weighted sum is $\widehat{\theta}_r - \theta$. The stated Lindeberg condition and nondegenerate variance limit give the two-sample triangular-array central limit theorem. Under a uniform $(2 + \delta)$ -moment bound, the RCT Lyapunov remainder is of order $n^{-\delta/2}$, and the OBS remainder also vanishes when $n/N \rightarrow \kappa < \infty$. The same moment control gives ratio consistency of the combined sample-variance estimator. Slutsky's theorem yields the studentized limit and Wald coverage.

B.15. Estimated density ratios and outer-fold honesty

For $\widehat{r}$ trained on the nuisance samples and then held fixed, subtract and add $P_O(rg) = P_R(g)$ to obtain the exact evaluation-sample drift

$$\mathbb{E}\{\widehat{\theta}_r(\lambda, \omega)\} - \theta = \omega P_O[\{\widehat{r}(X) - r(X)\}g(X)].$$

With $h_n = (\widehat{r} - r)g$, the second moment of the OBS evaluation perturbation is

$$\mathbb{E}\left[\left\{\sqrt{n}\omega(\mathbb{P}_{O,N} - P_O)h_n\right\}^2\right] = \omega^2 \frac{n}{N} \|h_n - P_O h_n\|_{L_2(P_O)}^2.$$

This proves the two negligible-drift requirements in Section 3; a nonnegligible ratio-training influence function or empirical remainder must instead enter the limiting influence function and its covariance.

For Proposition 6, condition on the prespecified or independently randomized split assignments and on one outer complement. Every fitted object is fixed, and the held-out paired folds retain the required RCT and OBS laws. The exact centering argument applies within that fold. Iterated expectation proves aggregate unbiasedness. It does not make different fold statistics independent: in general,

$$\text{Var}\left(\sum_k a_k \widehat{\theta}_k\right) = \sum_k a_k^2 \text{Var}(\widehat{\theta}_k) + 2 \sum_{k < \ell} a_k a_\ell \text{Cov}(\widehat{\theta}_k, \widehat{\theta}_\ell).$$

The second sum need not vanish because outer training complements overlap.

B.16. Counterexamples to adaptive reuse

The fixed-coefficient theorem cannot be extended by selecting a coefficient on the same evaluation outcomes without a separate analysis.

Reuse through λ . Let X be degenerate, $\theta = 0$, $Z_0 = 0$, and let the Δ_i be i.i.d. Rademacher variables. Every fixed λ satisfies $\mathbb{E}\{\lambda \mathbb{P}_{R,n} \Delta\} = 0$. If the same outcomes select $\hat{\lambda} = \text{sign}(\mathbb{P}_{R,n} \Delta)$, then

$$\mathbb{P}_{R,n}(Z_0 + \hat{\lambda} \Delta) = |\mathbb{P}_{R,n} \Delta|,$$

whose expectation is strictly positive.

Reuse through ω . Under the common covariate marginal, let $g(X)$ be a nondegenerate mean-zero Rademacher variable and let the RCT-valid score be identically zero. Every fixed ω centers $\omega\{\mathbb{P}_{O,N} g - \mathbb{P}_{R,n} g\}$. Choosing $\hat{\omega}$ as the sign of that same bracket produces its absolute value, which again has strictly positive expectation. The bias in both examples comes solely from adaptive reuse.